

NICOXIN 25% POWDER

Description and Composition

NICOXIN 25% POWDER is a yellow-to-yellow green powder anticoccidial premix powder. Each gram contains 250mg Nicarbazine.

Pharmacodynamics

Nicarbazine, a chemically synthesised product, is an equimolecular complex of 4,4'-dinitrocarbanilide (DNC) and 2-hydroxy-4,6-dimethylpyrimidine (HDP) of the formula $C_{19}H_{18}N_6O_6$. DNC, not HDP, has anticoccidial activity when used alone against *Eimeria*. The potency of DNC is increased ten fold when this molecule is complexed with HDP as nicarbazine, but no increase in anticoccidial activity is observed with a simple mixture of the two compounds (Cuckler *et al.*, 1955). Rogers *et al.* (1983) indicated that dissolution of the DNC-HDP complex in the gut results in the formation of much smaller DNC crystal size than if the water insoluble DNC is administered alone that would explain an increased efficacy. According to Porter and Gilfillan (1955), nicarbazine is at least partially broken down in the gut to DNC and HDP which may be absorbed independently. Based on some of the data obtained from clinical studies as well as from *in vitro* experiments (Ryley and Wilson, 1976), nicarbazine is capable to prevent coccidiosis. Cultures of *E. tenella* treated with nicarbazine at 0.5mg l⁻¹ culture medium for 72 hours starting 0 and 24 hours after inoculation slowed 1st generation schizont development and no 2nd generation schizonts were seen. Nicarbazine treatment at 1 or 2mg l⁻¹ culture medium led to fewer developmental forms being seen. Treatment starting 48 hours after inoculation had no effect on development of the parasites.

Pharmacokinetics

An excretion study was performed in chickens using nicarbazine (14C)-Radiolabelled both phenylurea and pyrimidinone portions of the molecule. The main excretion pathway for the pyrimidinone portion of the complex was in the urine (>90%). This demonstrated that this moiety was well absorbed. It was also rapidly eliminated and by the third day after the last dose, 83% of the pyrimidinone had been eliminated. By contrast, the phenylurea portion of the nicarbazine complex was predominantly excreted (90%) through faeces and at a slower rate than the pyrimidinone but the majority of the radioactivity was recovered in the first 3 days after withdrawal of medication. The observed urinary concentrations for the phenylurea portion were only 5 - 10% of those of the pyrimidinone indicating that kidney was not the major elimination pathway.

Indication

NICOXIN 25% Powder is used as an aid in the prevention of coccidiosis in chickens caused by *Eimeria acervulina*, *E. brunetti*, *E. maxima*, *E. necatrix* and *E. tenella*.

Recommended Dose

Prevention of coccidiosis: 500g per ton of feed.

Route of Administration

For oral administration via feed.

Contraindication

Do not use in known cases of hypersensitivity to active ingredient.

Warning and Precautions

Do not use as a treatment for coccidiosis.

Do not use in flushing mashers.

Do not feed to laying or breeding hens in production.

Treated birds must not be slaughtered for use in food for at least 4 days after the latest treatment with this drug.

Interaction with Other Medicaments

None known.

Pregnancy and Lactation

Do not feed to laying hens.

Symptoms and Treatment of Overdose

Nicarbazine should not be fed to layers, as it can cause discolouration and reduced hatchability of eggs (although the effect is reversible once the nicarbazine is withdrawn). It also may result in reduced heat tolerance in birds exposed to high temperature and humidity.

Storage Condition

Store below 30°C.

Shelf Life

3 years.

Withdrawal Period

5 days.

Packing

25kg.

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